

GEOMETRIC EARTHWORKS IN THE STATE OF ACRE, BRAZIL: EXCAVATIONS AT THE FAZENDA ATLÂNTICA AND QUINAUÁ SITES

Sanna Saunaluoma

Geometric earthworks located in the interfluvial zone of the Upper Purús and Madeira river tributaries in the southwestern Amazon are formed by ditches of varying shapes and sizes along with contiguous exterior embankments and roads terraced with low backfill banks. The earthworks in the Brazilian state of Acre were used from 1200 B.C. to the fourteenth century A.D., indicating a continuous, collective cultural institution, a generalized regional phenomenon characterized by local variants and alterations and probably not exclusively restricted to a specified ethnic group. The carefully planned position of the earthworks in the landscape and the recurring geometric forms represented in this earthwork architecture suggest functions that were part of a tradition of shared ideology related to rituals and/or the sociopolitical activities of ancient Amazonian peoples. Recent archaeological fieldwork on two composite earthwork sites, Fazenda Atlântica and Quinauá, situated in the core location of the geometric earthworks, provides further evidence for the predominantly ceremonial use of these constructions.

Las estructuras geométricas de tierra situadas en la zona interfluvial de los afluentes del Alto Purús y del Río Madeira en el suroeste de la Amazonia están formadas por zanjas de formas y tamaños diversos junto con terraplenes exteriores y caminos con bajos bancales de relleno. Las obras de tierra en el estado brasileño de Acre fueron utilizadas a partir de 1200 a.C. hasta el siglo catorce A.D., lo que indica una institución cultural colectiva, un fenómeno generalizado regional caracterizado por las variantes locales y probablemente no exclusivamente restringido a un grupo étnico específico. La ubicación cuidadosamente planificada de las obras de tierra en el paisaje y las formas geométricas recurrentes representadas en esta arquitectura de construcciones de tierra sugieren funciones que formaban parte de una tradición ideológica compartida relacionada con rituales y/o las actividades sociopolíticas llevadas a cabo por las poblaciones prehistóricas amazónicas. Las investigaciones arqueológicas realizadas recientemente en los sitios de Fazenda Atlântica y Quinauá, compuestos de obras de tierra circulares y cuadradas y situados en la área central de las estructuras geométricas de tierra, proporcionan más evidencia para el uso predominantemente ceremonial de estas construcciones.

Geometric earthworks in the Brazilian state of Acre have recently aroused the curiosity of scholars studying the Amazonian past. These monumental earthwork complexes are located in the interfluvial zone of the Upper Purús and Madeira river tributaries in the southwestern Amazon and are formed by ditches of different geometric shapes and varying sizes in association with contiguous exterior embankments (Dias 2006; Pärssinen et al. 2003; Pärssinen et al. 2009; Saunaluoma and Schaan 2012; Schaan et al. 2010; Schaan et al. 2012). The individual enclosures vary in diameter from dozens of meters to almost 400 m. The best-preserved ditches are still several meters deep.¹ In most cases, ancient roads terraced with low back-

fill banks function as entrances to the enclosures and connect the distinct geometric structures. Ondemar Dias carried out the first archaeological studies and investigations on earthwork sites in Acre under the auspices of the Programa Nacional de Pesquisas Arqueológicas na Bacia Amazônica project in the late 1970s.² Since then, ca. 300 individual geometric earthworks have been registered in eastern Acre, the westernmost division of the state of Rondônia, and the southern part of the state of Amazonas (Schaan et al. 2010). Comparable earthworks have also been recorded in the Bolivian Amazon in the departments of Pando and Beni (Arellano 2002; Arnold and Prettol 1988; Erickson 2010; Prumers et al. 2006; Saunaluoma 2010; Walker 2008a).

Sanna Saunaluoma ■ Institute for Cultural Research, Department of Archaeology, P.O. Box 59, FI-00014 University of Helsinki (sanna-kaissa.saunaluoma@helsinki.fi)

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Figure 1. Location of the geometric earthworks and main rivers in the state of Acre mentioned in the text (map courtesy of the Geoglifos da Amazônia Ocidental project).

The earthworks were used in Acre for a surprisingly long time, from 1200 B.C. to the fourteenth century A.D. (Dias 2006; Saunaluoma and Schaan 2012; Schaan et al. 2012; see also the dates discussed below), suggesting a continuous, collective regional tradition. Examples of similar pre-columbian regionality are seen in the central Llanos de Mojos of the Bolivian Amazon, where extensive raised fields, canals, and ditches are linked to adjacent mound and forest island settlement sites by the wide-ranging routes of causeways (Denevan 1966; Erickson 2001, 2006, 2010; Lombardo and Prümers 2010; Walker 2008b), and in the Upper Xingu region, where well-designed road networks connect clusters of late precolonial plaza villages across the terrain (Heckenberger 2005; Heckenberger et al. 2008). The regionally connected societies that once occupied these monumental domesticated landscapes were characterized by cultural diversity and dynamism, advanced political organization, and ritual and elite exchange (Heckenberger and Neves 2009).

The recent archaeological data presented here concern two composite earthwork sites situated in

the core area of the Acrean geometric earthworks and provide further evidence of the predominantly ceremonial use of these constructions. The structural features and layout of the sites and the distribution and characteristics of the cultural deposits within the enclosed areas point more toward their initial function as centers of public ritual and political affairs than toward permanent residential or economic utilization.

The Fazenda Atlântica and Quinauá Sites

In recent decades, drastic alterations in the Acrean landscape have transformed the former *terra firme* rain forest ecozone into open grasslands for cattle ranching and agriculture in many parts of the state. Paradoxically, the current devastation of the forest has contributed to the study of ancient Amazonian societies. The earthworks have become more exposed in deforested areas, particularly when seen from the air. Aerial photographs and satellite images have become effective tools for discovering and registering earthwork sites in Acre (Ranzi et al. 2007; Schaan et al. 2010). The earthworks are

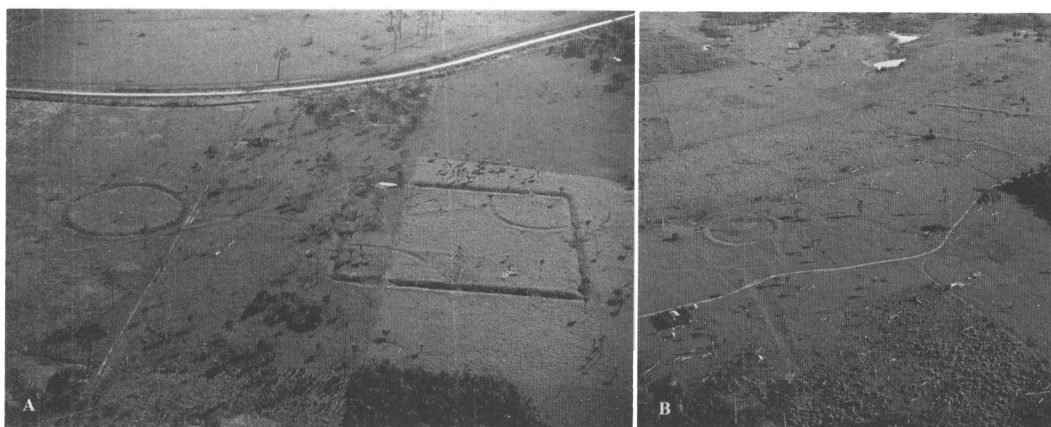


Figure 2. Aerial photographs of the (A) Fazenda Atlântica and (B) Quinauá sites.

situated on the terra firme plateaus of the Acre, Iquiri, and Abunã rivers, at approximately 180–230 m asl (Schaan et al. 2010) near rivulets. The distance between the separate earthwork sites varies, but generally other sites are located within a radius of 2–10 km.

The investigated sites Fazenda Atlântica (10°04'42"S 67°34'27"W) and Quinauá (10°03'24"S 67°37'01"W) are situated 5 km apart in the municipality of Senador Guimard some 35 km to the southeast of the city of Rio Branco, on nonflooded ground (ca. 200 m asl) of the drainage divide of the Acre and Iquiri rivers (Figure 1). The distance to the floodplain of the Acre River is ca. 15 km. The interfluvial plateau, pertaining to the western sedimentary upland (Sombroek 2000:Figure 1), which varies from undulating to flat and is marked by a web of numerous small headwater streams, represents the main landscape element in the research area. The soils in the region are mostly clayey tropical acrisols, with occasional layers of ferruginous laterite (Oliveira and Noquiera Alvarenga 1985). In Acre, the mean annual temperature is 24°C, and the mean annual rainfall is 2,100 mm, with a drier season extending from June to September (Teixeira 2008:70–73).

In 2007 mapping and measuring of the earthworks were conducted on-site using total station documentation.³ During the fieldwork period in 2008 ceramic and radiocarbon samples were collected from trial excavations carried out at the sites.

Fazenda Atlântica

The most prominent earthwork of the Fazenda Atlântica site (Figure 2A) is a 250-x-250-m square-

shaped ditch with shallow semicircular ditches inscribed within the western and eastern corners. Ditch construction backfill forms the 50-cm-high external embankment of the square. Bulldozing has partially altered the northern segment of the earthwork, but the preserved part of the ditch is currently approximately 3 m deep and 20 m wide. An eroded, 40-cm-high, roughly quadrangular embankment is located in the middle of the square-shaped ditched enclosure. A small artificial mound structure, 1.2 m high, 15 m in diameter, and attached to the internal embankment structure, is situated adjacent to the eastern semicircular ditch. A grid of auger units 50 cm in depth and 3 m apart from one other confirmed that the mound is formed by dark brown cultural soil limited to an area covering 15 x 15 m and therefore making up a separate feature. The main entrance of the earthwork, defined by a 100-m-wide road, is located on the southeast part of the square-shaped ditch. Minor roads, delineated by 40- to 50-cm-high earthen walls, begin at the northeast and southwest sides of the square-shaped ditch and disappear into the terrain approximately 100 m from the ditch. On the northwest side, two simple embankments parallel to the ditch in a northeast–southwest direction join and form a road leading to a circular ditch situated 150 m to the northwest of the square-shaped ditch. The circular ditch has a diameter of 125 m, its depth is approximately 1 m, and its width is 12 m. A 70-cm-high embankment surrounds the exterior of the circle (Figure 3).

In 2007, a test unit of 1 x 2 m was excavated in a north–south direction on the crest of the external embankment on the southwest side of the square-

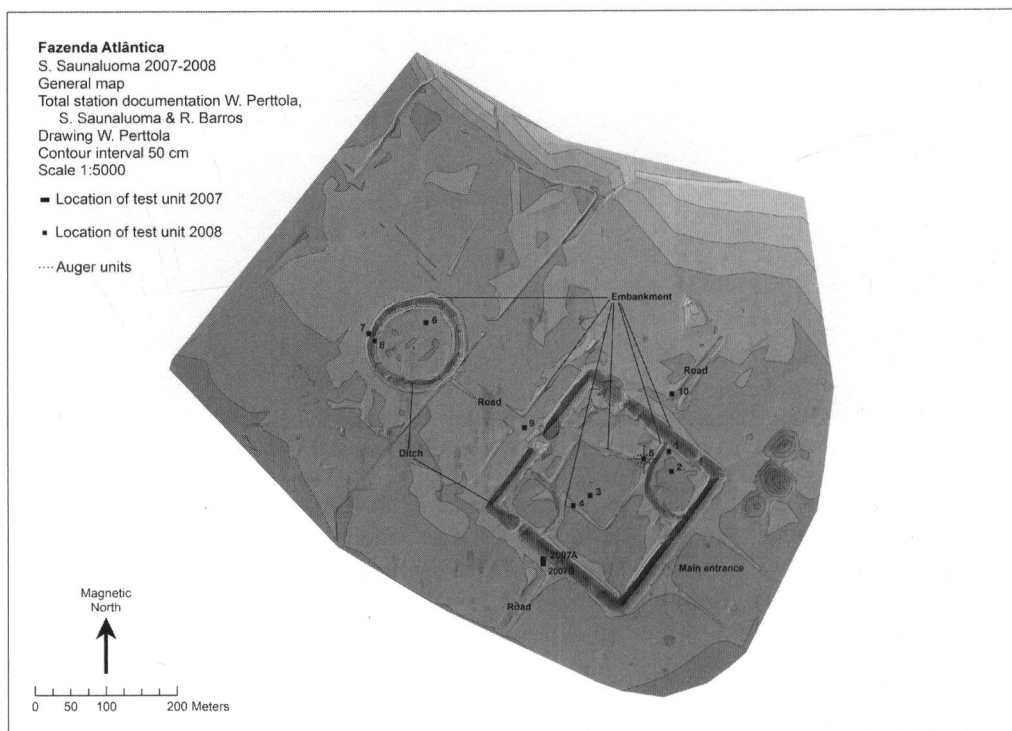


Figure 3. Contour map of the Fazenda Atlântica site.

shaped earthwork. Some scattered fragments of undecorated ceramics, charcoal, and carbonized palm seeds were found at a depth of 10–80 cm (Table 1), but the soil did not have any prominent coloring or stratigraphy. A small, simple, polished stone axe was collected from the outer slope of the ditch near the excavation.

In 2008, 10 1-x-1-m test units were excavated to artificial 10-cm levels. Units 1 and 2, placed inside the area marked by the semicircular ditch in the eastern corner of the earthwork, and Unit 3, roughly in the center of the square-shaped ditch inside the quadrangular embankment, revealed hardly any signs of cultural activities. Unit 4 was excavated near the western semicircular ditch on top of the embankment inside the square. A concentration of ceramics was found at the 10–40 cm level. The cultural layer also contained fragments of charcoal and burned clay. In spite of the considerable amount of archaeological material, the soil had the reddish tone typical of tropical latosols.

Unit 5 was placed on the small mound structure within the square-shaped ditch (Figure 4). Pottery, including carefully decorated sherds, was found in

the dark brown soil immediately under the humus layer. Furthermore, the structure of the soil was looser, and its texture was sandier, than in other areas of the site. Only the 10–20 cm level had a minor quantity and quality of ceramics; all of the other levels, until a depth of 130 cm, yielded a considerable amount of sherds. The major accumulation of pottery (1,060 sherds, 6.0 kg) was found in the 70–120 cm levels. The dark brown soil in these levels was mixed with charcoal. The 110–120 cm level revealed a feature that was approximately 20 x 60 cm in size and 2 cm thick (Figure 5). This feature was composed of a vessel broken in situ, fragments of charcoal and burned clay, chunks of calcium carbonate (1.1 kg), and tiny fragments of unburned bone.⁴ A clayey, reddish, culturally sterile soil was reached at a depth of 140 cm.

Units 6–8 were excavated adjacent to the circular ditch. Unit 6, placed inside the circle, suggested an extremely compact soil without any cultural remains. Unit 7 was placed on the crest of the external embankment on the northwest side of the earthwork. This unit exhibited a compacted soil mixed with fragments of ferruginous laterite and

Table 1. Distribution of Cultural Material in Excavated Units.

Unit	Location of Unit	Depth of Unit (cm)	Levels with Cultural Material (cm)	Number of Sherds Collected	Total Weight of Sherds Collected (g)
<i>Fazenda Atlântica</i>					
2007A	Embankment	100	10–80	180	685
2007B	Embankment	70	10–50	98	565
1	Square, interior	60	–	–	–
2	Square, interior	60	10–40	18	57
3	Square, interior	60	–	–	–
4	Embankment	80	0–60	491	2,665
5	Mound	150	0–130	1,665	9,500
6	Circle, interior	60	–	–	–
7	Embankment	80	10–60	51	178
8	Ditch base	60	–	–	–
9	Road	60	20–30	13	60
10	Embankment	100	40–80	291	905
<i>Quinauá</i>					
1	Enclosure, interior	60	0–20	7	35
2	Embankment	150	70–130	83	670
3	Ditch base	220	40–210	818	8,845
4	Circle, interior	60	–	–	–
5	Circle, interior	60	10–20	2	40
6	Circle, interior	90	10–70	393	6,690
7	Exterior	60	–	–	–
8	Road	60	–	–	–
9	Exterior	60	–	–	–
10	Road	60	20–40	4	70
11	Embankment	100	0–80	189	2,160
12	Road	60	–	–	–
13	Square, interior	60	–	–	–
14	Embankment	80	10–50	32	295
15	Exterior	60	–	–	–
16	Ditch base	250	30–230	145	3,046
17	Embankment	80	40–60	11	25
18	Exterior	60	–	–	–
19	Exterior	60	–	–	–
20	Embankment	80	10–30	19	735
21	Square, interior	60	10–20	19	220
22	Square, interior	60	0–20	11	155

occasional accumulations of eroded potsherds. Unit 8, excavated at the base of the circular ditch, had a recent fire layer in the 10–20 cm level but did not contain any archaeological material.

Unit 9, placed in the juncture of the embankments on the northwest side of the square-shaped ditch, yielded only a handful of potsherds. Unit 10 was excavated at the beginning of the road structure on the northeast side of the square-shaped ditch, on the crest of the elevated embankment some 30 m from the ditch. The first potsherds were observed at a depth of 40 cm, and the soil became loose in structure and darker in color with occa-

sional fragments of charcoal. An accumulation of sherds was noted at the 60–70 cm level.

Quinauá

The Quinauá earthwork site consists of a slightly inclined square-shaped ditch and a small circular ditch situated some 250 m to the south (Figure 2B). Both ditches have 70-cm-high external embankments. The sides of the square have a length of 100 m, and the diameter of the circular ditch is about 25 m. At present, the depth of the ditches is moderate (ca. 1.3 m), and the width is 8–10 m. The two earthworks are connected by a 20-m-wide road that

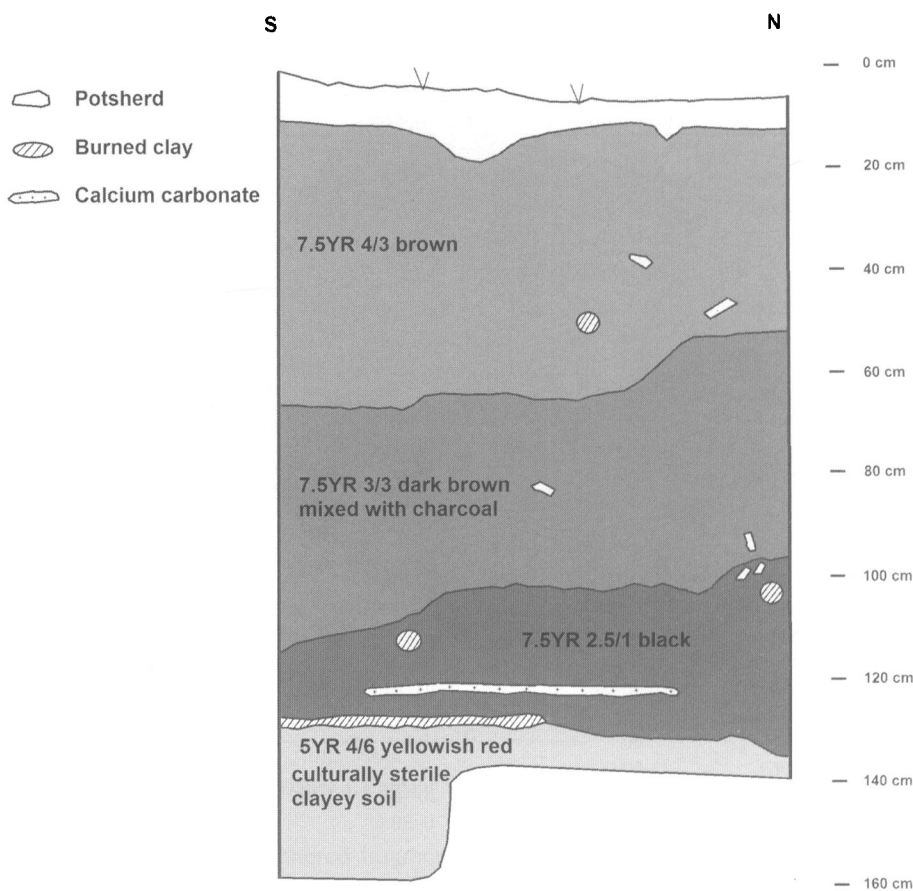


Figure 4. Western profile of Unit 5 at the Fazenda Atlântica site.

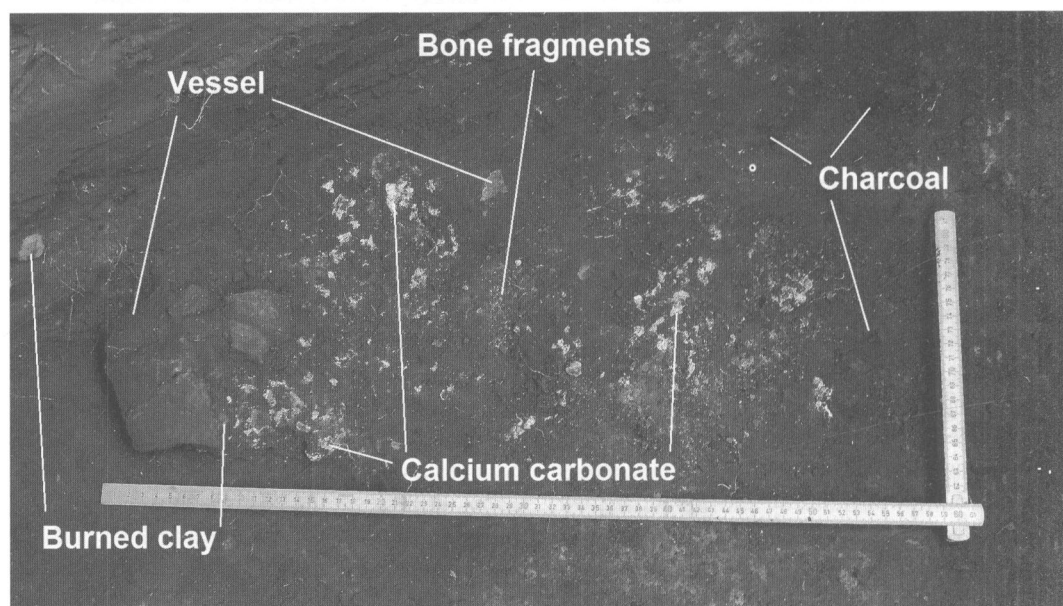


Figure 5. Unit 5, feature in the 110–120 cm level of the small mound at the Fazenda Atlântica site.

continues on the opposite side of the square toward a stream to the north. This road and elevated, 1-m-high parts on the external embankment mark the northern entrance of the square. On the eastern side of the square-shaped ditch we observed the remains of a 40-cm-high trapezoid-shaped embankment and a simple linear embankment in a north–south direction, situated 100 m from the ditch. A roughly square-shaped walled enclosure is connected to the southeast section of the circular ditch. The enclosure is approximately the same size as the circle and continues as a walled road some 200 m to the southeast in the direction of a stream (Figure 6).

In 2008, 22 units with dimensions of 1 x 1 m were excavated at the Quinauá site. Units 1–6 were placed near the circular ditch in the southern sector of the site. Inside the walled enclosure on the southeast side of the circle, Unit 1 yielded a few weathered fragments of pottery, but otherwise the compacted soil did not suggest any signs of cultural deposits. Unit 2, on the crest of the external embankment framing the circular ditch, had a moderate amount of potsherds at a depth of 70–130 cm. The structure of the soil in the first 50 cm of the excavation was loose with occasional fragments of charcoal. Unit 3, located at the base of the ditch, contained abundant pottery and some fragments of burned clay and charcoal. The major accumulation of cultural material was found in the 160–170 cm level. Unit 4, inside the circular earthwork, and Unit 5, on the inner margin of the circular ditch, did not suggest any archaeological deposits.

Unit 6 was excavated on the northern side of the circle, near the entrance where the road toward the square-shaped ditch begins. The sandy texture and reddish brown color of the soil observed in this sector differed considerably from those in other parts of the site. Robust utilitarian sherds were recovered from the cultural layer at a depth of 10–70 cm (Figure 7). At the level below where the last potsherds were found, the soil became compact, reddish, and culturally sterile.

Units 7 and 8 were placed halfway between the two earthworks, with Unit 7 some 20 m to the east of the road and Unit 8 in the middle of the road. Both units were culturally sterile, as was Unit 9, which was situated approximately 15 m south of the circular ditch. Unit 10, on the crest of the road wall on the southeast side of the circle, did not

show any indication of archaeological deposits either, excluding four small sherds.

Units 11–22 were placed close to the square-shaped earthwork. Unit 11 was excavated on an elevation of the external embankment on the north side of the site where the road structure enters the square. Most of the sherds collected from this unit were concentrated in the 30–70 cm levels. Unit 12, in the middle of the road 10 m to the north of the ditch, and Units 13, 15, 18, and 19, all placed within and between the earthen structures, did not contain any archaeological material. Unit 14, on top of the trapezoid-shaped embankment, and Unit 17, on the crest of the simple linear embankment, had small amounts of sherds. In addition to pottery, Unit 14 yielded some fragments of burned clay in the 60–70 cm level. Unit 16, at the base of the square-shaped ditch, had a small amount of sherds in the 30–40 cm and 50–70 cm levels, but a major accumulation of pottery and charcoal was found at a depth of 100–140 cm. Another smaller cluster of sherds was noted at a depth of 170–180 cm. Units 20–22, located on the exterior embankment and on the inner margins of the square-shaped ditch, contained a small amount of ceramics from immediately under the humus layer to a depth of 30 cm but no other signs of cultural activities.

Dating of the Sites

A basic issue with the geometric earthworks is their chronology, which is of the utmost importance for understanding the regional typology, the comprehensive extent of the occurrence of the earthwork sites, and their relationship to the cultural sequence of the Amazonian area in general. At the Fazenda Atlântica site we dated three different contexts (Table 2). The earliest date, 177 cal B.C.–cal A.D. 48, associated with the occurrence of ceramics, was obtained from the base of the cultural layer, roughly at the current ground level in the external embankment of the circular ditch. The embankment structure inside the square-shaped ditch is somewhat later, dating to cal A.D. 71–251. However, the existence of the road structure connecting the two earthworks suggests that they were probably used at the same time. The third date, cal A.D. 127–335, is from the feature in Unit 5, which included a shattered vessel, fragments of compacted calcium carbonate, and unburned bones (Figure 5). This suggests that the small mound of dark anthropic soil

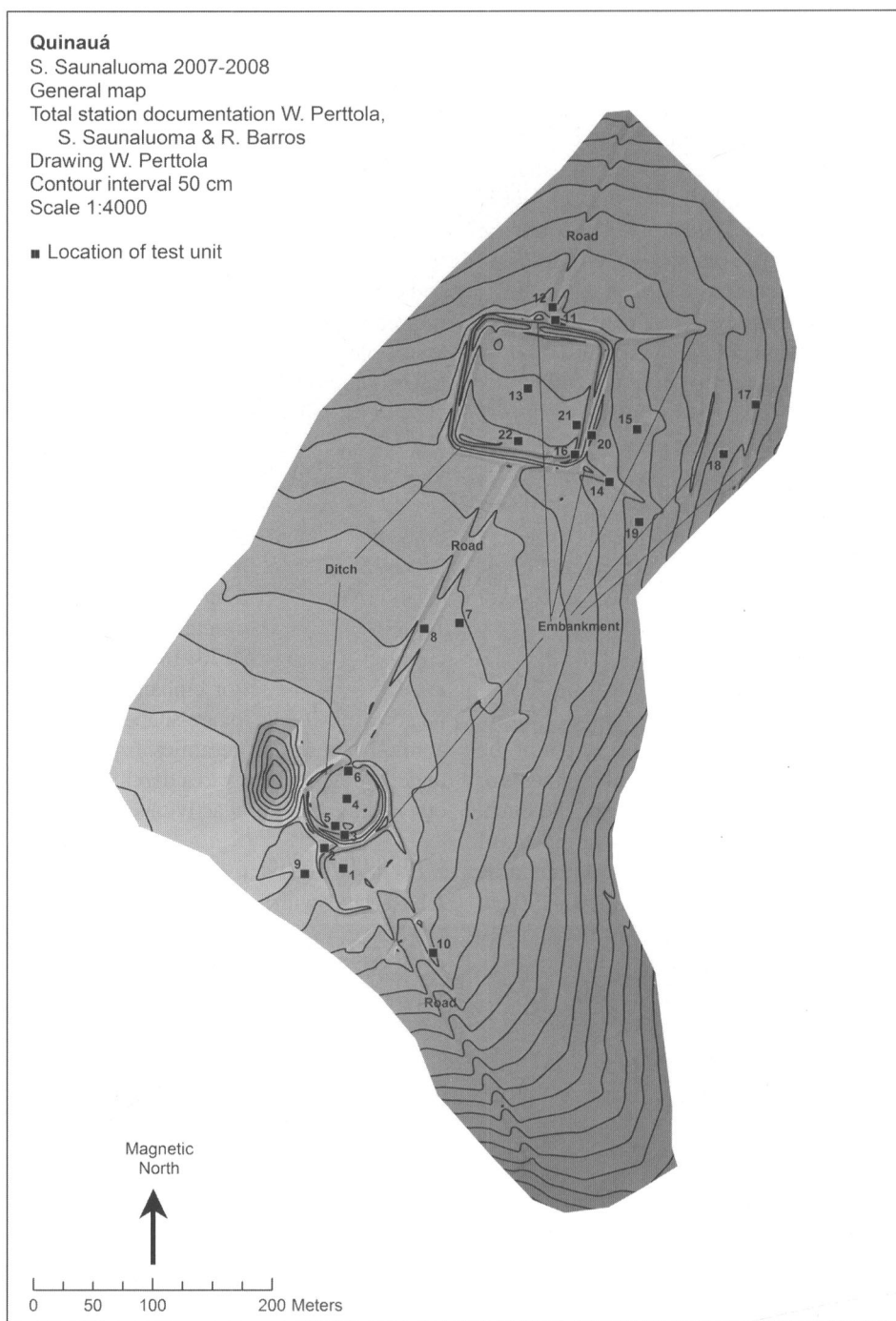


Figure 6. Contour map of the Quinauá site.

was possibly the most recent structure built, or formed, at the site.

The Quinauá site was occupied approximately a century later than the Fazenda Atlântica site. The

dates reveal that the deposit of darker cultural soil and abundant ceramics, situated in the northern section of the circular ditch, was noticeably earlier (cal A.D. 246–416) than the ditch structures. Char-

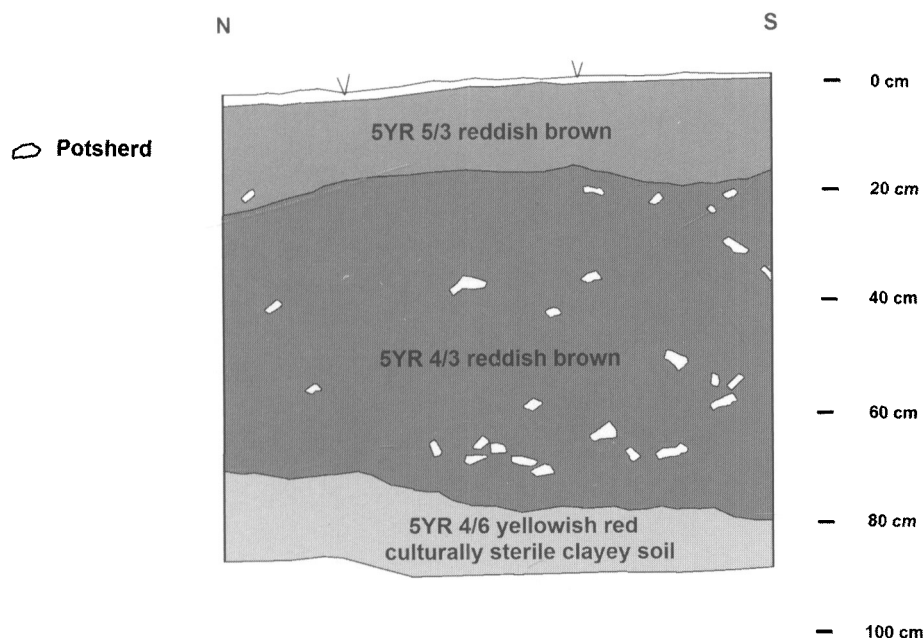


Figure 7. Eastern profile of Unit 6 at the Quinauá site.

coal recovered from cultural layers in the external embankment and at the base of the square-shaped ditch yielded dates of cal A.D. 426–592 and cal A.D. 429–594, respectively. The date cal A.D. 424–564 was obtained from the sooty surface of a sherd collected from the 160–170 cm level in Unit 3, placed at the base of the circular ditch. The dates indicate that the site was occupied before the construction and use of the earthworks and that both ditch structures were contemporaneous. Concurrent use of earthworks within the same site has also been noted at the Jaco Sá site. Two charcoal samples recovered from cultural layers at Jaco Sá, at the base of a simple square-shaped ditch and in the external embankment of another earthwork, a double-ditch structure, yielded dates of cal A.D.

780–977 and cal A.D. 782–984, respectively (Saunaluoma and Schaan 2012).

Thermoluminescence dates are available from Alto Alegre, a site, partly damaged by the BR-364 Highway, composed of a variety of road structures connected to a semicircular ditch attached to a circular ditch and situated only 2 km southwest of Fazenda Atlântica. These dates, obtained from Quinari-tradition ceramics, indicate a period of use at the site from about 200 B.C. to A.D. 200 (Nícoli 2000), contemporaneous with Fazenda Atlântica. The dates of the Fazenda Atlântica and Quinauá sites also coincide with the long utilization of the Fazenda Colorada earthwork complex. Charcoal recovered from the diverse cultural layers of the site yielded radiocarbon dates that place the occu-

Table 2. Radiocarbon Dates from the Fazenda Atlântica and Quinauá Sites.

Site	Provenience	Lab Number	$\delta^{13}\text{C}\text{‰}$ PDB	^{14}C Age (B.P.)	Calibrated Age (2 σ)
Quinauá	Unit 16, –230 cm, wood charcoal	Ua-37263	–26.9	1565 $\pm$ 35	A.D. 433–638
	Unit 11, –42 cm, wood charcoal	Ua-37262	–26.9	1570 $\pm$ 35	A.D. 432–623
	Unit 3, 160–170 cm, soot on sherd surface	Ua-37260	–13.1	1585 $\pm$ 30	A.D. 431–602
	Unit 6, –50 cm, wood charcoal	Ua-37261	–27.5	1760 $\pm$ 35	A.D. 246–416
Fazenda Atlântica	Unit 5, –110 cm, wood charcoal	Ua-37252	–25.6	1855 $\pm$ 30	A.D. 127–335
	Unit 4, –40 cm, wood charcoal	Ua-37251	–30.6	1905 $\pm$ 35	A.D. 71–251
	Unit 7, –55 cm, wood charcoal	Ua-37253	–27.2	2110 $\pm$ 35	177 B.C.–A.D. 48

Note: Dates were calibrated with the OxCal 4.1 program using the ShCal04 curve for the Southern Hemisphere.

pation of the Fazenda Colorada site between cal A.D. 69 and cal A.D. 1386 (Saunaluoma and Schaan 2012).

Ceramics

The Quinari ceramic tradition, as defined by Dias (2006; Dias and Carvalho 1988), is divided into four phases that can be related to the earthwork sites in the Upper Purus Basin. The Quinari tradition is characterized by a fusion of cylindrical and globular vessel forms with direct or everted rims, red slip as the primary surface-finishing technique, painted red and black lines on white slip and incised lines as the principal decorative elements, and the use of *caraipé* as the predominant temper, although charcoal, hematite, and grog tempers are also found (Dias 2006).⁵ Anthropomorphic vessels in the form of a cylinder on a globe representing a human face, executed using the appliqué technique, are, according to Dias (2006:190–191), a distinctive feature of the Quinari tradition.

The archaeological material excavated from the Fazenda Atlântica site during the 2007 field season consists of 278 potsherds (1.2 kg), of which 3 percent are diagnostic of the Quinari tradition. The sherds are fragmented and eroded to the extent that the incised lines on the few decorated pieces are almost indistinguishable. In 2008, 2,529 sherds (13.4 kg), of which 7 percent are categorized indisputably as Quinari ceramics, were recovered from the excavations. The pottery is fragmented, and the surfaces of the sherds are eroded, making it difficult to define the surface-finishing techniques and vessel body forms. The material excavated from Unit 5, placed on the small mound structure, is an exception, not only in quantity (making up 66 percent of all of the pottery recovered from Fazenda Atlântica) but also in the conservation state of the pottery, which is better. Altogether 82 percent (159 sherds) of finer decorated ware collected from the site was obtained from Unit 5.

In addition to potsherds, 18 fragments of laterite slabs were collected from Units 4, 5, and 10. Some of these laterite fragments have use-wear traces indicating their use as grinding stones (Figure 8). Ferruginous laterite is a raw material commonly found in the Acre region.

The nondecorated bulk ceramics of the Fazenda Atlântica site are generally poorly fired and have cores that are darker than the surfaces, as well as

some air bubbles. Decorated pottery is of a better quality, harder, and even in color. The coloring of the sherds varies considerably, ranging from pale gray to dark brown. The temper is predominantly organic, *caraipé* and charcoal, but occasional sand and hematite tempers were also observed. The surfaces are largely eroded, but some better-conserved sherds show brown, red, and creamy white slip, and some potsherds display brushed surfaces. The thickness of the vessel walls varies between 3 and 12 mm. The bases are mainly flattened, but one example of a rounded base was recovered. The rims are direct or slightly everted and sometimes thickened. Unit 5 also yielded some outward-folded rims. The predominate decorative technique is incision, but some painted sherds—red on white and red-and-black on white (Figure 9A)—and one fingernail-incised sherd with a flattened lip (Figure 9B) were recovered from Unit 5.

The design elements are geometric and curvilinear, and the design field is located on the upper part of the vessels. Parallel lines, incised on the inner side of the rim (Figure 10A), appear on many decorated sherds. In some cases, the lips of the rims are notched. Two molded pieces, that is, fragments of possible supports or lugs, and a spindle whorl fragment (Figure 9C) were found in the dark cultural layer in Unit 5. Several fragments (.8 kg in total) of a carinated bowl were recovered from Unit 5. When this vessel was reconstructed, a curious combination of incised linear, spiral, and meander designs, executed on the collar of the vessel and on the slightly rounded lip, was revealed (Figure 10B).

The archaeological material excavated from the Quinauá site consists of 1,813 potsherds (22.9 kg) and eight laterite slab fragments. Six of these lithic artifacts were found in Unit 6, in deposits suggesting an earlier settlement phase at the site. Only 4 percent of the pottery corresponds readily to the Quinari tradition, including mainly undecorated rims, although some incised body fragments and bases are also present. All recovered rims are direct with rounded lips. The bases are mostly slightly rounded and robust and are most likely fragments of large containers. Some of them are sooty inside. The pottery is tempered with charcoal and hematite particles, but *caraipé* and grog were also used. The surfaces of the ceramics are heavily eroded, but some sherds still have traces of brown and red slip. An example of finer ware—a fragment of a slightly

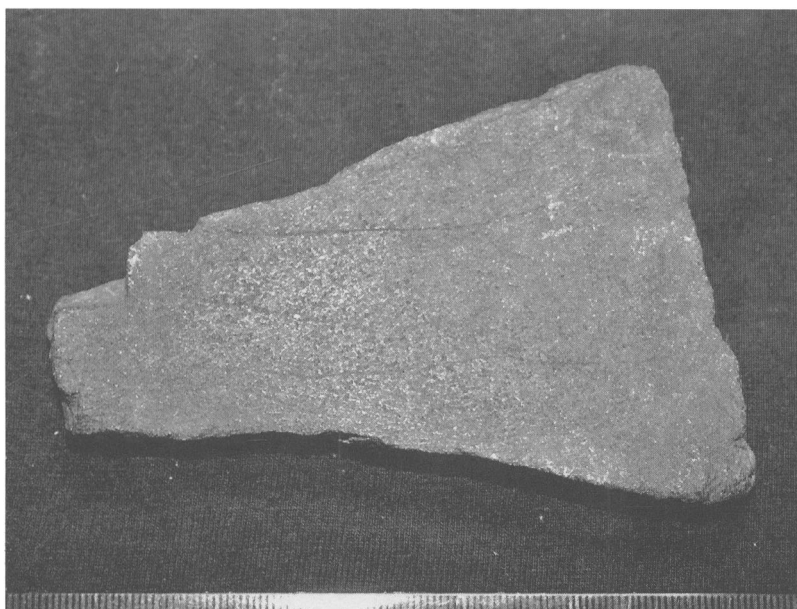


Figure 8. Grinding stone fragment recovered from the 100–110 cm level of Unit 5 at the Fazenda Atlântica site. Scale in millimeters.

angular lid that has an incised fine-line decoration on the exterior surface (Figure 9D)—was recovered from Unit 11.

Globular anthropomorphic vessels were not noted in the ceramic collections of the sites excavated during the present study, but the remaining type attributes of the Quinari tradition are present in the ceramic material of the Fazenda Atlântica and Quinauá sites. However, the fingernail incision and the carinated vessel form with a labial flange, present in the Fazenda Atlântica collection, are uncommon in the Quinari tradition. Carinated vessels are rather frequent in certain Amazonian ceramic complexes, such as the Incised Rim Tradition in the central Amazon (Lima 2008:273) and the Late Tutishcainyo (Lathrap 1970:90) and Yasuní pottery (Evans and Meggers 1968:11) in the Upper Amazon. The existence of different types of ceramics in the distinct mound context indicates a specific social relationship between diverse communities and may represent a sign of regional exchange focused on prestige building (Heckenberger 2005).

Archaeological Deposits and Structures

Unit 6, placed on the northern side of the Quinauá circular earthwork, at the juncture of the road connecting the ditch structures, exhibited a darkish

sand-textured cultural soil with abundant utilitarian pottery that mainly consisted of sooty robust fragments of larger vessels with some fragments of grinding stones and burned clay, suggesting, along with the radiocarbon dates, a settlement site preceding the construction of the earthworks at the location. The distribution of the archaeological material related to the earthworks was similar at both sites investigated during the present study, although landscape smoothing and depositional processes altered the earthworks more at Quinauá. At Fazenda Atlântica, the archaeological deposits were concentrated on the embankments and the small mound, while the flat surfaces enclosed by the ditches were practically empty of cultural layers and finds (Table 1). At Quinauá, in addition to those in the embankments, cultural deposits were concentrated in the ditches, while units excavated in areas inside and between the earthworks yielded scarce archaeological material. Unit 3, at the base of the circular ditch, suggested an accumulation of ceramics in the 160–170 cm level, perhaps indicating a more intensive phase of use at the site. The excavations proved that the ditches were originally much deeper than they are today. Although no test units were placed at the base of the square-shaped ditch of the Fazenda Atlântica site, ceramics are likely to be

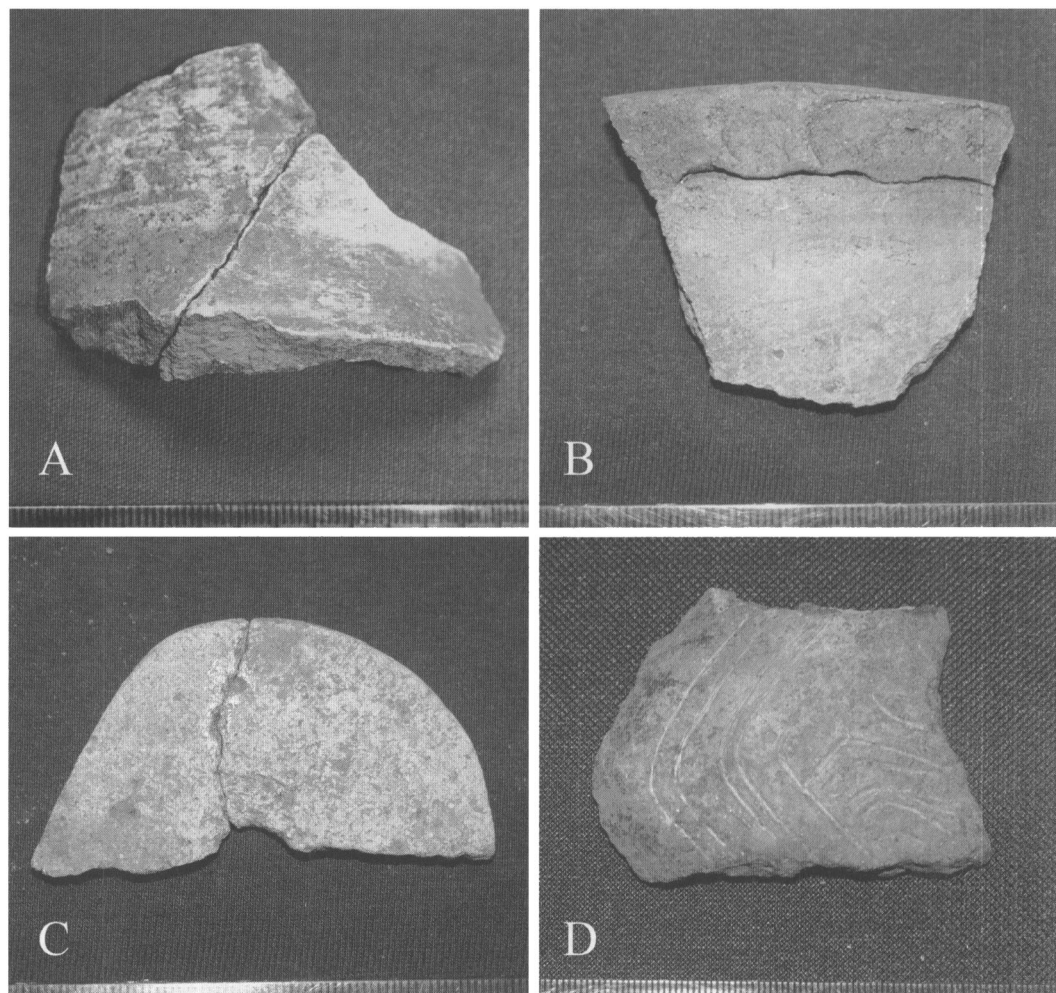


Figure 9. Potsherds recovered from (A–C) Unit 5 at the Fazenda Atlântica site and (D) Unit 11 at the Quinauá site: (A) direct rim sherd with a rounded lip from the 110–120 cm level, decorated with a red band and black geometric designs on a white exterior surface; (B) fingernail-incised direct rim sherd from the 100–110 cm level; (C) flat spindle whorl fragment from the 120–130 cm level; (D) fragment of a slightly angular lid from the 30–40 cm level. Scale in millimeters.

found in this context also, as frequent superficial finds were observed on the slopes of the ditch.

The fieldwork of Dias (2006) provided results similar to ours: The centers of the earthwork enclosures had few finds, and the cultural layers were found in the ditches and on the external embankments. Dias interprets the deposits of archaeological material in the ditches as a case of resilience, meaning that the ditches were inhabited sporadically during moments of crisis, both social and natural. However, it is likely that it would be impractical to dwell in a ditch several meters deep, primarily due to precipitation, which might cause

water to accumulate in the ditch. Naturally, the possibility of occasional camping in the ditches cannot be excluded, but then the question becomes whether the site was already abandoned and later reoccupied only for the short term, possibly by another group that had a conception of the significance, and perhaps the sacredness, of the earthworks different from that of the original builders of the site.

The geometric ditches were primarily signs of, and perhaps provided restricted access to, exclusive activity areas, such as spaces for ritual and/or political practices. As a secondary function, they

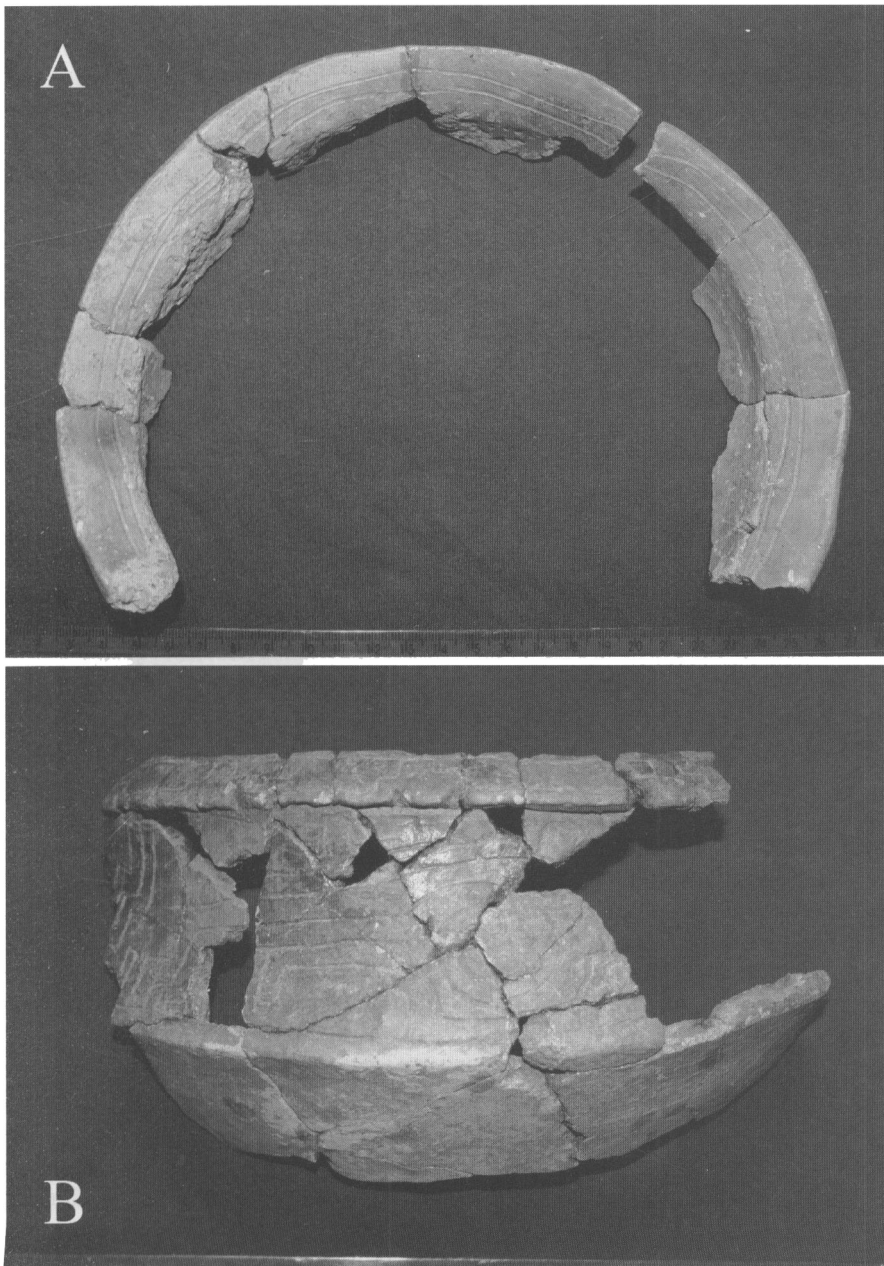


Figure 10. Pottery recovered from Unit 5 at the Fazenda Atlântica site: (A) rim sherds of an open-mouthed vessel from the 90–100 cm level; (B) fragmented carinated bowl with down-slanting labial flange from the 110–120 cm level.

might have served for waste disposal, which would explain the accumulation of potsherds and organic refuse in the base of the ditches. A similar patterning of cultural material has been noticed at some of the simpler enclosure sites of the northern Bolivian Amazon (Saunaluoma 2010). The easiest way to keep the public area of the site tidy would have

been to collect discarded material used during events that took place within the site from the interior of the enclosure and dump it into the ditch.

The ceremonial nature of certain geometric earthwork sites in Acre is indicated by the purposeful placement of finer ware in the ditches and elevated parts of the embankments near the

entrances. The ditch base of the main entrance at the JK site was a place for elaborately decorated pottery (Saunaluoma and Schaan 2012), implying that this context had a special, possibly a votive, function in the same manner as the small mound structure at Fazenda Atlântica. This mound of dark brown anthropic soil mixed with occasional fragments of charcoal is an especially interesting structure.⁶ The cultural layer in the mound contains a considerable quantity of pottery decorated with incision and painting, a pottery completely different from the utilitarian ceramics recovered from the other contexts of the Fazenda Atlântica and Quinauá sites. The feature in the 110–120 cm level consisting of a broken vessel, fragments of burned clay, cemented calcium carbonate nodules, and a few unburned bones is even more surprising. The explanation for this feature remains open, as does the function of the mound itself. If the process of the creation of the mound was semi-intentional, that is, a by-product of other activities, such as the accumulation of refuse, why were finer ware, calcium carbonate, and animal bones placed intentionally at the base of the mound?⁷ In any case, the small mound was a place to deposit special materials and not just conventional everyday commodities or debris.

Some of the mounds associated with ancient occupations in Amazonia contain burials (Machado 2005; Neves and Petersen 2006; Schaan 2007:51–53), but due to the reduced size of the trial excavation on the Fazenda Atlântica mound, the presence, or absence, of human remains and burial urns cannot be confirmed. In the central Amazon, artificial mounds bordering the plazas of certain precolonial villages and frequently including purposely destroyed vessels have been interpreted as habitation units initially constructed on elevated platforms (Moraes 2010) or as middens possibly associated with special activities linked to communal ritual and public occasions (Heckenberger et al. 1999; Machado 2005). The presence of the finer ware in the Fazenda Atlântica mound argues for the existence of prestige ceramics reserved for ceremonial use. The mound context could have functioned as an offering structure, exclusively utilized for certain public services that took place at the site.

The arrangement of semicircular ditches in the inner corners of a square is an attribute found so far only at the Fazenda Atlântica site, but some

other earthwork sites in Acre, such as California, Estrada Velha, Fazenda Nictheroy IV, Fazenda Vitoria, Nascentes do Quinauá, Nova Aldeia, Ramal dos Batalhas, and Rapirã, present a combination of a circular and a square-shaped ditch. Another shared structural feature observed at the Bela Vista, Chinésio, Dois Círculos, Fazenda Colorado, Garantã, and Xanadu I sites is a road leading to a walled enclosure connected to a circular ditch, such as at Quinauá.⁸ These sites are situated within a 50-km radius of the Fazenda Atlântica and Quinauá sites (Figure 1). The similarity of the structural patterns implies a close connection between the distinct sites, but without comprehensive chronological data it is still uncertain if they all were occupied roughly at the same time or consecutively, if the archaeological material related to these sites shares common characteristics, if there was any development in the layout or structural features of the sites, or if phases of reoccupation took place in the course of the cultural sequence of the geometric earthworks.

Earthworks Embedded in Landscape

Ditched earthworks related to prehistoric villages in the southern Amazon, in the Upper Xingu region (Heckenberger 2005; Heckenberger et al. 1999; Heckenberger et al. 2008), and in the eastern Bolivian Llanos de Mojos (Erickson 2010; Prümers et al. 2006) are interpreted as having a predominantly defensive function, although the earthen structures could have some symbolic, economic, and/or visual importance too. Heckenberger (2005:90) mentions that in the prehistoric defensive earthworks of the Xingu region the ditch construction overburden was always mounded on the interior edge of the ditch. At the Acrean sites the situation is the opposite. Generally the ditches have clearly marked exterior embankments, which casts doubt on the sole protective function of these structures.⁹ Walker (2008a) explains the ring-ditches of the central Llanos de Mojos as hydraulic systems, which were useful in the seasonally flooded savanna, and as boundary markers for the occupational zones. The former explanation is not applicable in the case of the Acre earthworks, as these are located on non-flooding ground. Throughout the period of their construction and use, when the ditches were deeper than they are today, they may have held water dur-

ing the rainy season, although it appears that they were not typically constructed to connect, change, or remove water flows.¹⁰ Walker's second interpretation is more likely regarding the earthen structures of Acre, and I propose that the geometric earthworks were markers of special places reserved for public sociopolitical and ritual practices.

The circular enclosure forms seem to predominate in the southern part of the area with geometric earthworks, while quadrangular forms are more common in the north (Schaan et al. 2007:Figure 4). In addition, the Acrean earthworks manifest themselves in other outline variants, such as ellipses, octagons, and U- and D-shapes. Nevertheless, I believe that for the ancient builders of the geometric earthworks, the circle, the square, and the straight line were the most important forms, and the other figures had a lesser significance in the earthwork architecture. The remains of material culture related to the earthworks are scarce and limited to imperfectly preserved pottery. However, the same symbols used in earthwork architecture may have also had a focal significance in other contexts, constructions, and objects.

If they were indeed constructed as public monuments, central spaces for varied gatherings and ceremonies, how do the geometric earthworks relate to ancient political and religious ideology? Staller (2008:2) asserts that "the landscape was culturally defined by ruling political and/or religious specialists from a certain place, usually major cultural centers that also had important symbolic significance to ethnic identity and social order." In the course of human history, it has often been important to transform new land into a given group's proper territory through the creation of a perceptible representation of the ideological values of the populace, as is done among the contemporary northern Arawakan groups in the South American tropical lowlands (Zucchi 2002), and to mark consecrated places physically in the landscape, as the construction of central monuments by Neolithic small-scale societies of the British Isles exemplifies (Tilley 1996). In many ritual and administrative centers of central Andean precolumbian societies, certain architectural elements, particularly the symmetry and balance of rectangular plazas and enclosures with (sunken) circular plazas, persisted throughout the centuries (e.g., Pozorski and Pozorski 2008; Shady 2006). The circle and

square forms probably also had a specific meaning in the cosmology of the ancient occupants of the Upper Purus Basin, which was reflected in the geometric earthwork architecture. In addition to the symbolic significance of the earthwork forms, the ditch structures functioned to delimit and restrict access to the special sections of the sites, while the enclosures and roads directed and regulated entry to and exit from the principal earthworks. As Tilley states (1996:168), "Circles imply motion within and around, lines motion along. Both types of monument demarcated spaces to cross, to go beyond, spaces to move into and out of, move between, look at and look beyond."

The consistency of the positioning and orientation of the earthworks in the landscape, the repetitive regularity of distinct earthen structures, and the layout of the sites argues for a certain cultural principle. Ethnohistoric sources contain some examples of the importance of circular and square forms for Amazonian peoples. Architectural monumentality, demonstrated in the circular plaza village system interconnected by extensive road networks, is considered typical of Arawakan groups (Heckenberger 2002, 2005). A circular village plan linked with socioceremonial organization is also common among the contemporary Gê of central Brazil (Wüst and Barreto 1999). During the initiation ceremonies of Apinayé warriors, a circular open area was cleaned at a distance of 200 m from the village and connected by a straight path to the easternmost point of the circle of houses (Nimuendajú 1983).¹¹ Alternatively, ethnographic literature refers to the rectangular communal huts and the octagonal ground plan of the male clubhouses among the Pano of the eastern Bolivian lowlands (Metrax 1942:42). However, spreading out in an area of diverse ecological and cultural settings in the southwestern Amazon, the geometric earthwork architecture may even have been shared beyond ethnic boundaries (Schaan et al. 2012). The geometric earthwork tradition can be regarded as a cultural tendency that was not necessarily restricted to a certain ethnic or linguistic group. It can be seen as a globalized regional phenomenon that was characterized by local variants and alterations.

The geometric earthworks were constructed on carefully selected elevated, yet level surfaces. Their location on interfluvial plateaus provided good visible control over the surrounding terrain, assuming

that vegetation was more or less as it is today, after a few decades of intensive logging of the forest, that is, more open and similar to a savanna.¹² Ethnoarchaeological study among the modern Kuikuru in the Upper Xingu region demonstrates that the villages and adjacent areas, located initially on forested terra firme, are swiftly converted to an environment of anthropogenically created savannas, parklands, low secondary forests, and cleared areas with recent gardens and occupation sites (Schmidt 2010). It is to be expected that the ancient societies inhabiting the terra firme plains in Acre also altered the environment by modifying and managing the natural world surrounding their occupational areas.

What distinguishes the geometric earthwork sites from many occupational sites in Amazonia is the lack of extensive anthropogenic *terra preta* (Amazonian dark earth) deposits. Amazonian dark earth is considered an indication of a continuing, fully sedentary agricultural way of life (Woods et al. 2009). Earthwork architecture conforms to the idea of a densely populated region with fixed long-lasting occupations, although not necessarily within the earthen structures or in their immediate vicinity. This means that the purpose of the geometric earthworks probably was not initially related to dwelling or agriculture. In any case, Acrean monumental earthwork architecture clearly contradicts the deep-rooted but presently withdrawing terra firme (upland)–*várzea* (floodplain) dichotomy, which made the Amazonian interfluvial zone incapable of sustaining widespread sedentary populations (e.g., Lathrap 1970; Meggers 1971). The distribution of the geometric earthworks roughly overlaps that of the natural stands of the Brazil nut (*Bertholletia excelsa* [Mori and Prance 1990:137]). Given the high nutritional value of the Brazil nut, it is likely that this is not a mere coincidence. Instead, it suggests that Brazil nut arboriculture may have been one of the sustaining resources of the earthwork builders.

Discussion

Earthwork engineering was a common practice throughout the Amazonian lowlands during the precolumbian period. The earthworks were built for different purposes, some more practical, some more symbolic, which is reflected in the diverse typological, functional, and structural categories that

present different cultural, regional, and temporal affiliations. Based on fieldwork carried out at the Fazenda Atlântica and Quinauá sites in Acre, the two geometric earthwork sites share common traits in their layout and features, the distribution of cultural deposits, and the attributes of the archaeological finds. The cultural material is concentrated mainly on the slopes and at the bases of the ditches and on the embankment structures, as if the fragments of pottery and burned clay were part of the construction fill for the embankments. For the most part, ceramic materials seem to be associated with the initial construction period or postprimary use of the sites. The amount of cultural activity remnants and the density of artifacts are low inside the area demarcated by the earthworks. It seems that these sections of the sites were kept clear of cultural debris and undergrowth. This was also the case for the roads. The test units excavated within these structures contained a scarcity of archaeological deposits. The ditch structures apparently protected but also controlled access to the special areas, while the roads guided passage in the direction of the central sectors of the sites.

Considering the distribution of archaeological material, the center of the earthworks was most probably a public area, that is, a plaza reserved for communal gatherings and socioceremonial activities, which is a cultural institution still in existence among some contemporary indigenous groups of the southern Amazon and central Brazil (Heckenberger 2005; Heckenberger et al. 2008; Nimuendajú 1983; Turner 2002; Wüst and Barreto 1999). The absence of dwelling deposits in the immediate vicinity of the earthworks is a distinction of the Acrean sites. The documented settlement layers at the Quinauá site seem to predate the construction and initial use of the earthworks. However, the Acre region still lacks systematic survey of habitation sites. One of the aims for future research is to identify the location, features, and intensity of the village sites related to the geometric earthworks and also to study the population's means of subsistence.

The carefully planned position of the earthworks in the landscape and the recurring geometric forms represented in earthwork architecture suggest functions that were part of a tradition of shared collective ideology related to the cosmology and/or sociopolitical concerns of the ancient peoples. The small mound structure located inside the square-

shaped ditch at Fazenda Atlântica implies a ritual use for the site. The carinated vessel recovered from the mound differs from the typical Quinari-tradition pottery, which could imply the existence and importance of an elite ware related to the exchange of symbolic prestige goods in a larger interlinked regional system (Heckenberger 2005): “Ritual performance in highly structured public ceremonial spaces and material culture were primary mechanisms of social communication within multiethnic and, in some cases, multilingual regional sociopolitical systems” (Heckenberger and Neves 2009:259, citing Barreto 2008 and Lathrap 1985).

The monumental earthworks in Acre were discovered only fairly recently, but the ditches and embankments are already filling in and eroding, primarily due to present-day land use. The meaning and magnitude of a precolumbian earthwork phenomenon of a genuine Amazonian origin are just beginning to be understood, but the possible disappearance of these sites may be only decades away. In addition to studying them, we should also find a way to protect these unique testimonials of the Amazonian past.

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- Meggers and Clifford Evans of the Smithsonian Institution. For a more comprehensive general description of the archaeological research realized in Acre, see Schaán 2008.
3. The total station measurements were taken with a 600-series Geodimeter. On average, 1,320 measurements were made on each site. Given the considerable area of the sites and the available time per site, the parts with flat or simple topography were mapped sparsely, while the earthworks were measured in more detail. This means that structures that were not noticed in the field are unlikely to be visible in the surface models. The maps are based on our perception of the sites, and there might be more structures and features waiting to be found.
4. Some of the unburned bones were identified as fragments of the upper jaw and teeth of agouti (family *Dasyproctidae*), a rodent with widespread distribution in the Amazonian area. The rest of the bones were of unidentified species and may have been from larger mammals and birds.
5. *Carapê* is siliceous tree bark (*Licania* sp.) ash.
6. Mounds are found at some earthwork sites in Acre, for example, at Coqueiral, Sol de Maio (Schaán and Bueno 2008), Fazenda Iquiri II (Schaán and Bueno 2009), and Campo Esperança (Schaán and Saunaluoma 2010), but in these cases several mounds form a circular enclosure, and therefore their function is probably different from that of the individual structure documented at the Fazenda Atlântica site.
7. Excavations at the Las Palmeras earthwork site in the Bolivian Amazon (Saunaluoma 2010) revealed that the small depression features located inside a circular ditch were filled with abundant sherds and midden debris. In addition, we observed a calcite feature in one of the depressions at a depth of 70–80 cm, but the implication of this feature remains uncertain. Today, calcite has more uses than almost any other mineral as a construction material, a pigment for paint, a field dressing, and a medication.
8. At the Fazenda Colorada site the structure linked to the enclosure is a U-shaped double ditch instead of a circle.
9. The defensive function of small circular earthworks of the Adena culture in the North American Eastern Woodlands has been discarded precisely due to external piling of ditch construction backfill (Clay 1987).
10. The JK site, consisting of two concentric square-shaped ditches, was constructed in the margin of a stream so that the watercourse actually forms part of the outer ditch. However, this type of solution is uncommon in the earthwork architecture of Acre.
11. This pertains to the Macro-Gê linguistic family.
12. One of the objectives of the ongoing studies on the Acrean earthworks is to investigate the vegetation history of the region.

Notes

1. Soil creep and slope wash processes that transport the fine-grained sediments will fill up, widen, and flatten the earthworks noticeably over time (O'Neal et al. 2005). Today, the forms and dimensions of the earthworks, as well as the distribution of materials, may be quite different from the original ones.
2. Programa Nacional de Pesquisas Arqueológicas na Bacia Amazônica was a research project coordinated by Betty

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